University course pack

Teaching embedded and IoT systems at university

A 14-week, 3-credit module in embedded and IoT systems for third-year electrical, computer or mechatronics engineering students. Prerequisites: circuits, C programming and signals. The four Darsi lectures anchor weeks 3, 6, 8 and 9.

Learning outcomes

1. Choose sampling rates and ADC resolution and justify them from signal properties.
2. Predict and prevent aliasing in a measurement chain.
3. Build and measure an energy budget for a duty-cycled node.
4. Analyse schedulability of periodic tasks.
5. Select and justify a link and protocol for an IoT design brief.

14-week syllabus

Week	Topic	In class and out
1	Microcontrollers and GPIO	Lab 1: working code, one change
2	Sensors and signal conditioning	Lab 2: a sensor on the bench
3	Sampling, Nyquist and the ADC	Darsi lecture 1; aliasing lab
4	Timers and interrupts	Lab 3: debounced button; problem set 1
5	Real-time scheduling	RTOS tasks lab
6	Energy budgets and low power	Darsi lecture 2; current measurement lab
7	Wireless links	Range test outdoors
8	MQTT, CoAP and LoRa	Darsi lecture 3; broker in the room
9	Capstone design and safety	Darsi lecture 4; proposal due
10	Security for IoT devices	Seminar; quiz
11	Capstone build 1	Requirements and test plan
12	Capstone build 2	Midpoint demo; peer review
13	Capstone build 3	Measured power budget due
14	Final demos	Demo, defence and peer assessment

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Lecture slide outline

Lecture 1: Sampling and the ADC

- Hook: 12 beats per minute
- Sampling theorem fs > 2B
- Folding and anti-aliasing filters
- Quantisation step and SNR
- Predict, then measure

Lecture 2: Real-time and power

- Hook: the nine-day battery
- I_avg and duty cycles
- Where energy goes
- Hard and soft real-time
- Rate-monotonic bound

Lecture 3: IoT protocols

- Hook: the farm brief
- Publish/subscribe and brokers
- MQTT QoS 0, 1, 2
- CoAP over UDP
- Range against rate

Lecture 4: Capstone design

- Scaffold, then fade
- The lab sequence
- Safety and cost
- Engineering evidence
- Peer assessment

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Problem set

1. An ECG has useful content up to 150 Hz. Choose a sampling rate and anti-aliasing cut-off, and justify them.

2. A 130 Hz tone is sampled at 100 Hz. At what frequency does it appear?

3. How many bits are needed to resolve 1 mV over a 0 to 3.3 V range?

4. A node wakes for 0.5 s every 30 s at 24 mA and sleeps at 5 µA. Estimate life on 2,400 mAh.

5. Tasks (C, T) = (1, 5), (2, 10), (3, 20) ms. Is the set schedulable under rate-monotonic priorities by the Liu and Layland test?

6. For a valve-control command over MQTT, which QoS would you choose and why?

Solutions

> **1.** Nyquist needs > 300 Hz; choose about 500 Hz to 1 kHz with a cut-off near 150 to 200 Hz so the filter has room to roll off below fs/2.

> **2.** |130 - 100| = 30 Hz.

> **3.** 3.3 V / 1 mV = 3,300 levels; 2^{12} = 4,096 ≥ 3,300, so 12 bits.

> **4.** I_avg ≈ 24 × 0.5/30 + 0.005 ≈ 0.405 mA; 2,400 / 0.405 ≈ 5,900 h ≈ 8 months, before derating.

> **5.** U = 0.2 + 0.2 + 0.15 = 0.55 ≤ 3(2^(1/3) - 1) ≈ 0.78, so yes.

> **6.** QoS 1 with an idempotent command, or QoS 2 if duplicates are harmful; a lost command is unacceptable, unlike a lost reading.

Lab handout: A measured power budget

Goal

Predict a node's battery life from measurements and check the prediction.

Steps

1. Insert a 10 Ω shunt in the supply line or use a current meter; record active, radio and sleep currents.
2. Measure each phase's duration with a GPIO toggle and an oscilloscope or logic analyser.
3. Compute I_avg and predicted life with a 20% derating margin.
4. Change one design choice (wake interval, payload or sleep mode) and recompute.
5. Run the node on a small cell for 48 hours and compare charge used with the prediction.

Hand in

The measurement table, the budget, the comparison and a one-page explanation of any gap.

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Assessment rubric

Criterion	Excellent	Good	Developing	Not yet
Measurement	Accurate, repeated, uncertainty stated	Accurate	Partly measured	Estimated only
Engineering analysis	Correct and linked to design choices	Correct	Minor errors	Major errors
Requirements and tests	Every requirement tested	Most tested	Some tested	No test log
Safety and practice	Exemplary	Safe	Minor lapses	Unsafe

Reading list

Lee, E. A., & Seshia, S. A. (2017). Introduction to Embedded Systems: A Cyber-Physical Systems Approach (2nd ed.). MIT Press.
Core text: sensing, timing and scheduling.

Shannon, C. E. (1949). Communication in the presence of noise. Proceedings of the IRE, 37(1), 10–21.
The sampling theorem.

Liu, C. L., & Layland, J. W. (1973). Scheduling algorithms for multiprogramming in a hard-real-time environment. Journal of the ACM, 20(1), 46–61.
The rate-monotonic bound.

Atzori, L., Iera, A., & Morabito, G. (2010). The Internet of Things: A survey. Computer Networks, 54(15), 2787–2805.
IoT architectures.

OASIS (2019). MQTT Version 5.0. OASIS Standard.
The MQTT specification.

Shelby, Z., Hartke, K., & Bormann, C. (2014). The Constrained Application Protocol (CoAP). RFC 7252, IETF.
The CoAP specification.

Biggs, J., & Tang, C. (2011). Teaching for Quality Learning at University (4th ed.). Open University Press.
Constructive alignment.

Freeman, S., et al. (2014). Active learning increases student performance in science, engineering, and mathematics. PNAS, 111(23), 8410–8415.
Evidence for active learning.